

Yared Teweldebirhan Weldu

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ABOUT ME

Aspiring AI Engineer with strong Python and machine learning skills, focused on building AI and IoT solutions for real problems. I led Hiwet AI, a student project selected as a finalist for the Zayed Sustainability Prize that secured a \$25,000 grant for the school, and represented Ethiopia at the award ceremony in Abu Dhabi. Experienced in leadership, teaching, and building working prototypes from AI models to mobile app front end, back end, and database.

PROJECTS AND LEADERSHIP EXPERIENCE

2024/2025

Team Leader – Hiwet AI: Disease Detection for Indigenous Crops

[Kallamino Special High School, Tigray Development Association, Ethiopia](#)

- Co-proposed an AI mobile app that detects diseases in teff, sorghum, barley and cactus pear from a photo, offline and with Tigrigna support.
- Planned a CNN built from scratch with TensorFlow, Keras and Google Colab, trained on real local images, plus a school committee to train 100+ students.
- Submission selected as a finalist in the Zayed Sustainability Prize, securing a USD 25,000 grant to expand the initiative; presented at the award ceremony in Abu Dhabi, UAE.

2025/2026

Developed – Hiwet Smart Poultry Management System

[Kallamino STEM Center](#)

Built a prototype robot and Android app that detect poultry diseases (e.g. Newcastle disease, coccidiosis) from images and give recommended actions.

- Trained a MobileNet-based deep learning model and exported it to ONNX for on-device use in an Android Studio app with Amharic and English support.
- Robot built with Arduino Uno, ESP32 Wi-Fi module and DC motors; streams a live camera feed and monitors temperature and humidity to control a fan and lamps.

July – August 2025

Volunteer Tutor

[Voluntary Summer Tutorial Program, Tigray Development Association with the Tigray Regional Education Bureau](#)

Tutored students in Grades 8 to 12; recognised with a Certificate of Recognition for active participation, dedication and volunteerism.

2025/2026

Active member – Kallamino STEM Center

[Kallamino Special High School, Ethiopia](#)

Contributed to several significant STEM projects during the academic year.

2024/2025

Sport Club co-founder and President and Art Club member

[Kallamino Special High School, Ethiopia](#)

Active and voluntary participation in the Sport Club; valued contributions to artistic projects in the Art Club.

EDUCATION AND TRAINING

2024 – 2026

High school

[Kallamino Special High School, Tigray, Ethiopia](#)

[Fitawerary Abayeneh General Secondary School, Addis Ababa, Ethiopia](#)

May 2025	Android Developer Fundamentals – Nanodegree Udacity (verified certificate)
18 – 24 August 2025	TAIK AI Camp 2025– Capstone Project Winner The Africa I Know Developed an AI project addressing an African challenge.
Summer camps	INSA Summer Camp: Hosted by Ethiopian <i>Information Network Security Agency (INSA)</i> . AI Summer Camp: Hosted by <i>Ethiopian Artificial Intelligence Institute (EALL)</i> . Ethiopian Gifted and Talented Education Summer Camp

AWARDS AND ACHIEVEMENTS

2024/2025	Finalist, Zayed Sustainability Prize International competition; the Hiwet AI project earned a USD 25,000 grant for the school.
November 2025	3rd place, Ethiopian National Science Fair Ministry of Education, Ethiopia (held at Addis Ababa Science and Technology University) The competition was among students of STEM centers across the nation. Also received a participation certificate for the fair.
August 2025	Capstone Project Winner – TAIK AI Camp 2025 The Africa I Know
12 – 14 November 2025	Participant, Ethio–Israel Innovation Week Ministry of Innovation and Technology, Israel Embassy, STEMpower and Unistream, at AASTU

PERSONAL SKILLS

Mother tongue	Tigrigna
Other languages	Amharic, English, Arabic
Digital and technical skills	- Python and machine learning (TensorFlow, Keras, CNN) and NLP (Transformers) - Android app development (Android Studio) - Microcontrollers and IoT (Arduino Uno, ESP32), prototype robotics
Communication and leadership	Team leadership (Team Leader of an internationally recognised project), teaching and tutoring, presenting at national and international events.
Other skills	Speed typing, swimming, riding a bicycle and a motorbike

INTERESTS

Hobbies	Reading books, playing football
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